

G. S. Mandal's Maharashtra Institute of Technology, Aurangabad (An Autonomous Institute) Department of Computer Science and Engineering						
MID SEMESTER EXAMINATION-II Academic Year 2023-24 Semester-I						
Class: TYCSE Course: OS Max Marks: 15				Date: Time: 1 hr		
Q. 1	Solve/Answer any two of the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	Enlist objectives of File Management.	2	3	2	1	1.3.1
b)	Define Relocation	2	3	2	2	2.3.1
C)	Describe File Management Function.	2	4	2	4	4.3.1
Q. 2	Solve/Answer Any one					
a)	Describe Disk Space Management in detail.	5	3	3	3	3.1.1
b)	Compare Windows and Unix File System.	5	3	3	3	3.1.1
Q. 3	Solve/Answer Any one					
a)	Explain in detail File Allocation Method.	6	4	4	4	4.1.1
b)	What are the Memory Management requirements explain in detail.	6	4	4	4	4.1.1

Course Coordinator

Programme Coordinator

G. S. Mandal's
Maharashtra Institute of Technology, Aurangabad
(An Autonomous Institute)
Department of Computer Science and Engineering

MID SEMESTER EXAMINATION-I

Academic Year 2023-24 Semester-I

Class: TYCSE

Course: OS

Max Marks: 15

Date: 21/09/2023

Time: 1 hr

Q. 1	Solve/Answer any two of the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	What is the primary role of an Operating System as a user/computer interface?	2	1	2	1	1.3.1
b)	Define multiprogramming in the context of OS.	2	1	2	1	1.3.1
c)	Define Process Management.	2	1	2	1	1.3.1
Q. 2	Solve/Answer Any one					
a)	Explain the concept of multitasking Operating System.	5	2	3	2	3.1.1
b)	Write a short note on CPU Scheduling	5	2	3	2	3.1.1
Q. 3	Solve/Answer Any one					
a)	Explain in detail Time Sharing Operating System with advantages, disadvantages, and example	6	2	3	2	5.1.1
b)	Discuss the five-state process model.	6	2	4	2	5.1.1

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END SEMESTER EXAMINATION

Third Year B.Tech (CSE) – Dec/Jan-2023-24

Course Code: CSE302

Course Name: Operating System

Duration: 2 Hrs

Max. Marks: 50

Date: 16/12/2023

Instructions:

- All questions are compulsory
- Assume suitable data wherever necessary and clearly state it
- Figures to right indicate full marks

Q. 1 Answer any five (Marks:10)

	Marks	CO	BL	PI
a) Define System Call.	2	CO1	2	
b) What are the functions of operating system.	2	CO1	2	
c) Differentiate between user level and kernel level threads.	2	CO2	1	
d) Draw the diagram of sequential access method of file management.	2	CO3	2	
e) What is meant by Paging.	2	CO4	2	
f) Define RAID.	2	CO5	2	
g) How to prevent deadlock.	2	CO6	2	
h) What is Operating System evaluation.	2	CO1	2	

Q.2 Discover Producer-Consumer problem in detail.**OR(optional)**

Illustrate Dining Philosopher Problem Using Semaphores.

Q.3 Illustrate linked File Allocation with diagram.**OR(optional)**

Discuss different Types of File Sharing.

Q.4 Discuss in detail the Allocation of frame in OS.**OR(optional)**

Discuss the concept of segmentation in detail.

Q.5 Describe in detail disk scheduling algorithm.**OR(optional)**

Describe in detail disk formatting for device management.

Q.6 Apply deadlock avoidance for single and multiple resources.**OR(optional)**

Distinguish between Linux and windows OS

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END SEMESTER EXAMINATION

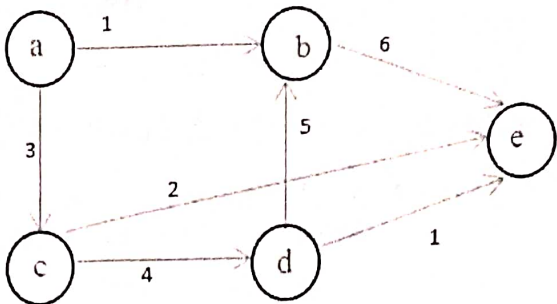
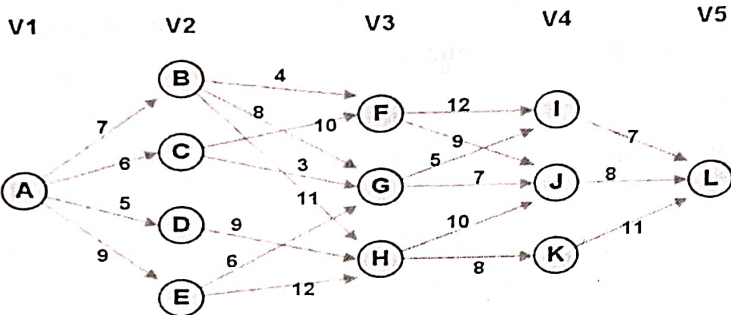
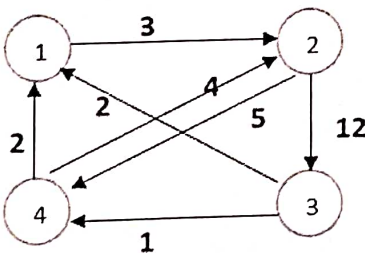
B.Tech. Third Year (CSE/AIDS) Dec/Jan-2023-24

Course Code: CSE301/AIDS302 Course Name: Design & Analysis of Algorithms
Duration: 2 Hrs Max. Marks: 50 Date: 14/12/2023

Instructions:

- i) All questions are compulsory
- ii) Assume suitable data wherever necessary and clearly state it
- iii) Figures to right indicate full marks

Q.1	Answer any five(Marks:10)	Marks	CO	BL	PI
a)	What is performance analysis? Explain the basic parameters used for performance analysis.	2	1	2	2.2.3
b)	Explain the asymptotic notations used for representing algorithm complexity.	2	1	2	2.2.3
c)	Apply binary search to find key = 45 in the given list of numbers: 12, 32, 45, 54, 59, 60, 68, 75, 80, 91, 95	2	2	3	3.2.2
d)	What is Quick Sort? How does it work?	2	2	2	2.2.3
e)	Find an optimal merge pattern for the 4 files whose lengths are: 35, 4, 40, 15	2	3	3	3.2.2
f)	Explain 0/1 knapsack problem.	2	4	2	2.2.3
g)	What is Traveling Sales Person problem?	2	5	1	2.2.3
h)	What is Clique problem?	2	6	1	
Q.2	A) Sort the following list using Merge Sort: 15, 67, 45, 36, 88, 92	4	1	3	3.2.2
	B) Solve the following recurrence relation with substitution method and find the time complexity: $T(n) = \begin{cases} 1 & n = 1 \\ 2T\left(\frac{n}{2}\right) + n & n > 1 \end{cases}$	4	2	4	
Q.3	A) What is Fractional Knapsack problem? Find an optimal solution to the fractional knapsack instance n = 7 with capacity of knapsack m = 15: (p1:p7) = (10, 5, 15, 7, 6, 18, 3) (w1:w7) = (2, 3, 5, 7, 1, 4, 1) OR B) Find the shortest path from vertex (a) to remaining	8	3	3	3.2.2

	vertices for the diagram using Single Source Shortest Path algorithm. 				
Q.4	A) Find a minimum cost path from s (vertex A) to t (vertex L) in the following multi stage graph:  OR B) Find the minimum distance between every vertex of the following graph using all pair shortest path algorithm: 	8	4	3	3.2.2
Q.5	A) Explain the 4-Queen's problem and write its implicit and explicit constraints. Solve the 4-Queens problem with backtracking. OR B) Explain the 15-puzzle problem with example. Draw a portion of space tree for the problem.	8	5	4	3.2.2
Q.6	Write short note on any two of the following: - NP-Complete and NP-Hard problems - Cook's Theorem 3 SAT problem	8	6	2	2.2.3

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MID SEMESTER EXAMINATION-I

Academic Year 2023-24 Semester-I

Class: TY (CSE)

Date: 11/09/2023 .

Course: Design & Analysis of Algorithms

Time: 1 hr

Max Marks: 15

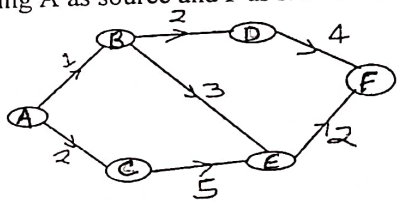
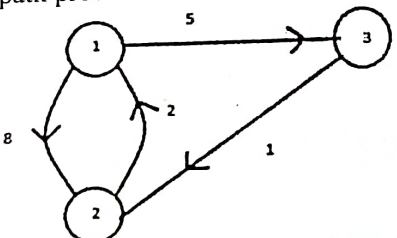
Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	What is an algorithm? What are the properties of an algorithm?	2	1	1	1	
b)	What is performance analysis of algorithm? What are the basic parameters for algorithm analysis?	2	1	2	1	
Q. 2	Solve/Answer Any one					
a)	Solve the following recurrence relation with substitution method and find the time complexity: $T(n) = \begin{cases} 1 & n = 0 \\ T(n-1) - n & n > 0 \end{cases}$	5	1	4	3	
b)	Solve the following recurrence relation with Master's theorem and find the time complexity: $T(n) = 4T\left(\frac{n}{2}\right) + n^2 \log^2 n$	5	1	4	3	
Q. 3	Solve/Answer Any one					
a)	Sort the following list with Merge Sort: 12, 33, 27, 40, 5, 19, 55, 47	6	2	3	5	
b)	What is Binary Search? Write recurrence relation of Binary Search and calculate its time complexity.	6	2	4	2	

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MID SEMESTER EXAMINATION-II
Academic Year 2023-24 Semester-I
Question Paper

Class: TY CSE
 Course: Design & Analysis of
 Algorithms
 Max Marks: 15

Date: 14/10/23
 Time: 1 hr

Q. 1	Solve/Answer the following questions	Marks	CO	BL	PI
	Questions	2	04	02	
a)	What is 0/1 knapsack problem?	2	03	02	
b)	What is general greedy approach for problem solving?				
Q. 2	Solve/Answer Any one				
a)	Explain activity selection problem with example.	5	03	03	
b)	Explain Job sequencing with deadline problem with example.	5	03	03	
Q. 3	Solve/Answer Any one				
a)	Solve the following multistage graph problem considering A as source and F as sink vertex: 	6	04	02	
b)	Consider the following graph and solve the all pair shortest path problem: 	6	04	02	

Bharti
 Course Teacher

S.G.
 Module Coordinator

S.G.
 Program Coordinator

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Maharashtra Institute of Technology, Chh. Sambhajinagar

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MAKEUP/SUPPLY EXAMINATION

Third Year B.Tech (CSE) – June/July-2024

Course Code: CSE304

Course Name: Formal Languages and Automata Theory

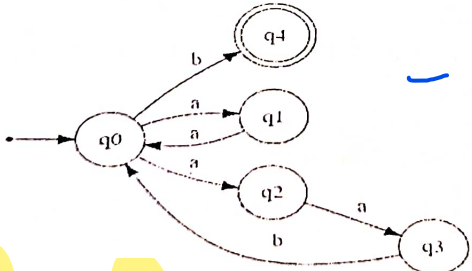
Duration: 2 Hrs

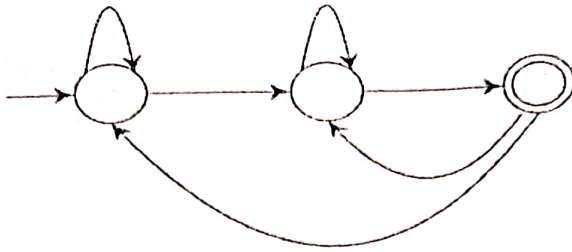
Max. Marks: 50

Date: 20/07/2024

Instructions :

- All questions are compulsory
- Assume suitable data wherever necessary and clearly state it
- Figures to right indicate full marks

Q. 1 Answer any five(Marks:10)					
		Marks	CO	BL	PI
a)	Draw and explain model of Turing Machine.	2	CO4	L1	1.4.1
b)	What is pumping lemma for FA?	2	CO2	L2	
c)	Define: (i) Finite Automaton(FA) (ii) Transition diagram	2	CO1	L1	
d)	Enlist applications of CFL	2	CO3	L2	
e)	Explain Chomsky hierarchy?	2	CO3	L2	
f)	What is recursive and recursively enumerable language?	2	CO4	L2	
g)	Define Deterministic PDA.	2	CO3	L1	
h)	Differentiate between Mealy and Moore Machine.	2	CO2	L3	
Q.2	Convert following NFA to DFA. Draw Transition diagram for equivalent DFA and Write Initial state and Final state.	4 4	CO1	L3	
 OR State whether following transition diagram is DFA or NFA. Why? Draw transition table for the same and check acceptability of strings: 011010 & 1101101		8			



Q.3 Construct a DFA with reduced states equivalent to the R.E. $\{10 + (0 + 11)0^*1\}$.

OR(optional)

Enlist and explain various operators of Regular Expressions with suitable example. Represent the language, defined over $\Sigma = \{a,b\}$ using Regular Expression also draw DFA for the same.

1. Having at least one 'a'
2. Starting with a and ending in b OR starting with b and ending in a

Q.4 Explain Ambiguity in grammar. Check whether the grammar $\{E \rightarrow I, E \rightarrow E+E, E \rightarrow E^*E, E \rightarrow (E), I \rightarrow a | b | c\}$ $w = \{(a+b^*c)\}$ is ambiguous or not?

OR(optional)

Reduce the following grammar to GNF: $\{S \rightarrow XB | AA, A \rightarrow a | SA, B \rightarrow b, X \rightarrow a\}$

Q.5 Construct PDA accepting all palindromes over (a,b)

OR(optional)

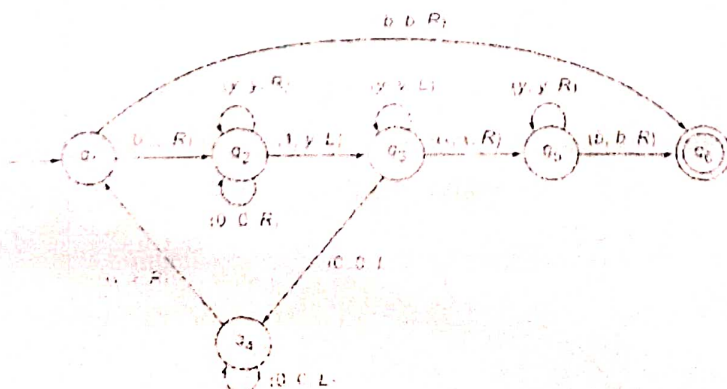
Construct PDA for the given CFG, and test whether the string $\{010000\}$ is acceptable by this PDA. $\{S \rightarrow 0BB, B \rightarrow 0S | 1S | 0\}$

Q.6 Explain following with suitable example.

- a. Post Correspondence Problem.
- b. Halting Problem of Turing Machine

OR(optional)

For the following Turing Machine, draw the computations for strings: 1. $\{00\}$ and $\{0011\}$ also comment on acceptability of these strings.



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G. S. Mandal's Maharashtra Institute of Technology, Aurangabad (An Autonomous Institute) Department of CSE						
MID SEMESTER EXAMINATION-II Academic Year 2023-24 Semester-I						
Class: TY(CSE) Course: Formal Languages & Automata Theory Max Marks: 15				Date: Time: 1 hr		
Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	What is Context-Free Grammar? Write derivation for string 0000111. $S \rightarrow 0S A$, $A \rightarrow 0A 1B$, $B \rightarrow 1$	2	2	2	2	1.4.1
b)	Whether the given grammar is in CNF or not? Give reason. a) $S \rightarrow ABC BC CD$, $A \rightarrow a$, $B \rightarrow b$, $C \rightarrow c$, $D \rightarrow a$ b) $S \rightarrow AB$, $S \rightarrow BC d a b c$	2	3	2	4	
Q. 2	Solve/Answer Any one					
a)	Let $G = (\{S, A, B\}, \{a, b\}, S, \{S \rightarrow aA, A \rightarrow aA \mid B, B \rightarrow \lambda \mid bB\})$. Find $L(G)$.	5	2	4	4	
b)	Define Push Down Automata. Explain Acceptance by PDA.	5	3	3	3	
Q. 3	Solve/Answer Any one					
a)	Reduce the following grammar into Chomsky-Normal Form $G = \{S \rightarrow 1A \mid 0B, A \rightarrow 1AA \mid 0S \mid 0, B \rightarrow 0BB \mid 1S \mid 1\}$	6	3	4	3	
b)	Construct a PDA 'A' accepting the set of all strings over $\{a, b\}$ with equal number of a's and b's.	6	3	4	3	

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MID SEMESTER EXAMINATION-I

Academic Year 2023-24 Semester-I

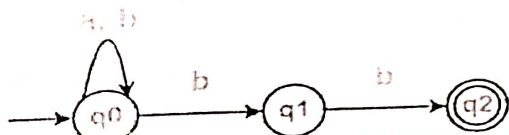
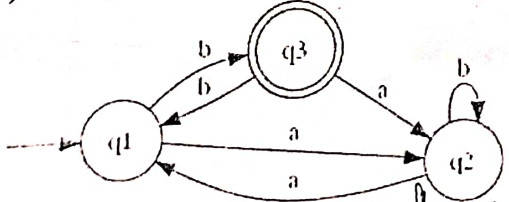
Class: TY(CSE)

Date: 04/09/23

Course: FLAT

Time: 1 hr

Max Marks: 15

Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	State True or False. Also clarify your answer. "Moore m/c can be transformed to FA."	2	CO1	4	2	1.4.1
b)	Define Finite State Machine.	2	CO1	1	2	
Q. 2	Solve/Answer Any one					
a)	Describe with example Finite Automata with output.	5	CO1	2	3	
b)	What is Regular expression? Give RE for representing set of strings over (0,1), i) Where every 0 is immediately followed by two 1's. ii) Containing all possible combinations of 0's and 1's but not having consecutive 0's	5	CO1	4	1	
Q. 3	Solve/Answer Any one					
a)	Construct a deterministic finite automaton equivalent to following NFA. Represent with transition diagram and transition table. 	6	CO2	3	3	
b)	Determine Initial and final state/s, check and describe whether the string is acceptable or not i) aaa, ii) bbabab iii) abbab 	6	CO2	3	2	

Mrs. Swati Jaipurkar
Course Teacher

Ms. Bharti Gurav
Module Coordinator

Dr. Smita Kasar
Program Coordinator

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Department of Computer Science & Engineering

MID SEMESTER EXAMINATION-II

Academic Year 2023-24 Semester-I

Class: TY

Course: SPPM

Max Marks: 15

Date:

Time: 1 hr

Q. 1 Solve/Answer the following questions						
	Questions	Marks	CO	BL	PO	PI
a)	In the context of the case study, for the following scenarios identify the most appropriate requirements elicitation techniques. 1. UI design of e-banking portal, ATM interface, computer systems 2. Analysis for understanding mode of transactions Checks, Cash, DD, MT, Gold, etc .	2	CO3	2	PO1	2.1.2
b)	Map the following project management activities to the respective project management phases. Justify the answer. - Updating the schedule plan due to addition of people to the project - Memorandum of Understanding (MoU)	2	CO4	5	PO2	3.1.1
Q. 2 Solve/Answer Any one						
a)	According to Boehm, what are the 3 stages of software cost estimation? Read the given scenario and find out the solution. A complex inventory management system of size 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the Effort, development time, average staff size, and productivity of the project.	5	CO3	3	PO4, PO5, PO9, PO11	2.2.4
b)	A project is estimated to be 400 KLOC. Calculate the effort and development time for each of the three model i.e., organic, semi-detached & embedded.	5	CO3	3	PO4, PO5, PO9, PO11	2.2.4
Q. 3 Solve/Answer Any one						
a)	Develop a plan for Recovery of a project from failure. Consider any project scenario.	6	CO4	6	PO4, PO5, PO9, PO11	3.1.2
b)	Summarize types of leadership style/Theory of leadership.	6	CO4	4	PO4, PO5, PO9, PO11	3.1.2/ 10.1.1

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Department of Computer Science & Engineering

MID SEMESTER EXAMINATION-I

Academic Year 2023-24 Semester-I

Class: TY
 Course: SPPM
 Max Marks: 15

Date: 21/09/2023
 Time: 1 hr

Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	Why Software engineering?	2	CO1	2	PO1	2.1.2
b)	Enlist the characteristics of SRS	2	CO2	2	PO2	3.1.1
Q. 2	Solve/Answer Any one					
a)	Analyze the given scenario to identify suitable process model for the development of the software. Write your observations. Justify why that model is suitable? Scenario: Customer has provided you all the requirements and has assured that there will not be any change in the requirements. He expects the deliverables from you at every stage of development. You have carried out a similar project earlier and you are sure that the project could be executed systematically.	5	CO1	4	PO1, PO2	2.2.4
b)	Compare between LSM and Prototype model	5	CO1	4	PO1, PO2	2.2.4
Q. 3	Solve/Answer Any one					
a)	Draw Use case diagram for the given scenario and summarize how the user and system will interact. Scenario: The web Customer makes use of any online shopping website to purchase online. The top-level uses are as follows; View Items, Make Purchase, Checkout, Client Register. Both View Recommended Item and Add to Wish List include the Customer Authentication, as they necessitate authenticated customers, and simultaneously item can be added to the shopping cart without any user authentication.	6	CO2	3/4	PO2, PO3	3.1.2
b)	Summarize the different categories of requirements with example of each.	6	CO2	4	PO2, PO3	3.1.2/ 10.1.1

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MID SEMESTER EXAMINATION-I

Academic Year 2023-24 Semester-I

Class: TY

Date:

Course: EEF&C

Time: 1 hr

Max Marks: 15

Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	What is Life Cycle cost?	2	1	2	1	1.2.1
b)	What is the Total Cost?	2	1	2	1	1.2.1
Q. 2	Solve/Answer Any one					
a)	What is Investment Analysis? Explain its different approaches and factors to be considered.	5	3	3	4	4.1.2
b)	Explain the Break Even Analysis with neat sketch	5	3	3	4	4.1.2
Q. 3	Solve/Answer Any one					
a)	What is cost estimation? Explain the different methods of cost estimation.	6	3	3	4	4.1.2
b)	Explain the Life cycle cost.	6	3	3	4	4.1.2

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MID SEMESTER EXAMINATION-I

Academic Year 2023-24 Semester-I

Class: TY (CSE)

Date: 1/09/2023 .

Course: EEF&C

Time: 1 hr

Max Marks: 15

Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	Write down the formula to calculate the compound interest.	2	1	2	1	1.1.1
b)	What is net cash flow?	2	1	2	1	1.3.1
Q. 2	Solve/Answer Any one					
a)	Explain the economics of growth and development.	5	2	3	1	1.2.1
b)	Explain the concept of Present value and Future value.	5	2	3	1	1.1.1
Q. 3	Solve/Answer Any one					
a)	Explain the importance and scope of economics in engineering	6	1	3	1	1.2.1
b)	Explain cash flow with classification.	6	1	3	1	1.2.1

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END SEMESTER EXAMINATION

Third Year B.Tech. (All) Dec/Jan-2023

Course Code: HSM301

Course Name: Engineering Economics, Finance & Costing

Duration: 2 Hrs

Max. Marks: 50

Date: 11/12/2023

Instructions:

- i) All questions are compulsory
- ii) Assume suitable data wherever necessary and clearly state it
- iii) Figures to right indicate full marks

Q. 1 Answer any five(Marks:10)					
		Marks	CO	BL	PI
a)	Define the term Economics.	2	1	2	
b)	What do you mean by interest rate?	2	1	2	
c)	What do you understand by opportunity cost?	2	2	2	
d)	State the Law of Demand.	2	2	2	
e)	What do you understand by Direct cost and Indirect cost?	2	3	1	
f)	What do you mean by Life Cycle Costing?	2	4	1	
g)	Enlist the causes of Depreciation.	2	4	2	
h)	Define Direct and Indirect Taxes.	2	2	2	
Q.2	Distinguish between Micro and Macro Economics. OR Calculate the value of deposit of Rs. 10000 made today for 6 years. If the Interest rate is, I) 9% if compounded Annually. II) 10% if compounded Half Yearly. III) 12% if compounded Quarterly. IV) 5% if compounded Monthly.	8	1	3	
Q.3	Write a note on cash flow diagram. OR A Machine costs Rs. 65,000/- to purchase. Its annual operating and maintenance costs are Rs. 3000/-. Machine saves Rs. 12,000/- in labour cost. The salvage value of Machine after 5 years is Rs. 6000/- and disposal cost is Rs. 500/-. 1. Draw the cash flow diagram. 2. Calculate the net cash flow.	8	3	2	

Q.4	Explain the Break Even Analysis with neat sketch. OR A Machine is purchased for Rs. 1,00,000. If after 5 years, its scrap value is estimated as Rs. 25,000. Calculate: 1. The Annual depreciation 2. Rate of Depreciation and 3. Book value at the end of every year in tabular form by using Straight line method.	8	5	2	
Q.5	What do you understand by Insurance? What are its types? Explain in detail. OR Suppose you are having Rs. 3,00,000 surplus fund with you and you wish to invest the same. You have options to invest in any Two of the following financial instruments. Which will you choose? And Why? Justify your Investment with reasons. 1. Fixed deposit (FD) 2. Equity shares 3. Mutual Fund (Equity/Debt) 4. Gold ETF 5. Post Office MIS 6. Public Provident Fund 7. National Pension Schemes (NPS) 8. Bonds 9. Debentures 10. Treasury Bills	8	4	3	
Q.6	What is Share Market? How does it function? OR Explain GST in detail.	8	6	2	

Note: - All course outcomes shall be addressed.

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END SEMESTER EXAMINATION - June/July-2023-24**B.TECH VI Semester (CSE)**

Course Code: CSE354

Duration: 2 Hours

Max. Marks: 50

Course Name: Cloud Computing

Date: 08 June 2024

Important Instructions:

1. All questions are compulsory.
2. Assume suitable data wherever necessary and clearly state it.
3. Figures to right indicate full marks.

Question 1: Answer any 5 questions.

Marks CO

- | | | | |
|----|---|---|---|
| a. | Explain the cloud ecosystem? | 2 | 1 |
| b. | Define cloud computing and identify its core features? | 2 | 1 |
| c. | What are the disadvantages of virtualization? | 2 | 2 |
| d. | What are the most important advantages of cloud technologies for social networking application? | 2 | 2 |
| e. | List the names of some popular software-as-a-service solutions. | 2 | 3 |
| f. | What are the characteristics of cloud architecture? | 2 | 4 |
| g. | What are the innovative characteristics of cloud computing? | 2 | 5 |
| h. | What are the objectives of cloud information security? | 2 | 6 |

Question 2: Answer any 2 questions.

Marks CO

- | | | | |
|----|--|---|---|
| a. | What do you mean by scalability? Explain in detail. | 4 | 2 |
| b. | Compare Private cloud versus public cloud. | 4 | 2 |
| c. | What is community cloud? Explain the scenario in which it could be suitable. | 4 | 2 |

Question 3: Answer any 2 questions.

Marks CO

- | | | | |
|----|---|---|---|
| a. | List cloud enabling technologies Explain any two in details. | 4 | 3 |
| b. | Define virtualization? write various types of virtualization in detail? | 4 | 3 |
| c. | Explain virtualization techniques in cloud computing | 4 | 3 |

Question 4: Answer any 2 questions.

Marks CO

- | | | | |
|----|--|---|---|
| a. | What are basic requirements of a secure cloud software? | 4 | 4 |
| b. | Explain the importance of data encryption, access controls, and data loss prevention mechanisms in ensuring data security in the cloud | 4 | 4 |
| c. | What are the cloud security challenges? | 4 | 4 |

Question 5: Answer any 1 questions.

Marks CO

- | | | | |
|----|---|---|---|
| a. | Describe the major features of the Aneka Application Model. | 8 | 5 |
| b. | Describe how cloud computing technologies can be applied to support remote ECG monitoring | 8 | 5 |

Question 6: Answer any 1 questions.

Marks CO

- | | | | |
|----|---|---|---|
| a. | Discuss the core concepts of Microsoft Azure in cloud computing, highlighting their significance, key features, and use cases. | 8 | 6 |
| b. | Discuss the various compute services offered by Amazon Web Services (AWS) in cloud computing, highlighting their key features, use cases, and advantages. | 8 | 6 |

—End of question paper—

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Department of Computer Science and Engineering

MID SEMESTER EXAMINATION-II
Academic Year 2023-24 Semester-II

Class: TY CSE

Course: CNS

Max Marks: 15

Date: 06/04/2024

Time: 1 hr

Q. 1		Solve/Answer the following questions				
	Questions	Marks	CO	BL	PO	PI
a)	Distinguish between stream ciphers and block ciphers?	2	3	2	2,3,6,8	
b)	How can the Diffie-Hellman key exchange protocol be vulnerable to man-in-the-middle attacks?	2	4	3	2,3,6,8	
Q. 2		Solve/Answer Any one				
a)	How does the Advanced Encryption Standard (AES) algorithm ensure robust security in modern cryptographic systems, and what are the key factors contributing to its resistance against various cryptographic attacks?	5	3	4	2,3,6,8	
b)	Explain the role of the Feistel network structure in the Data Encryption Standard (DES) algorithm and how it contributes to both the security and efficiency of DES encryption.	5	3	4	2,3,6,8	
Q. 3		Solve/Answer Any one				
a)	Consider the plain text alphabet 'G'. Using RSA algorithm and the values as $P=17$ and $Q=29$ find N , E , D and the cipher text in an RSA encryption process.	6	4	4	2,3,6,8	
b)	How does the Diffie-Hellman key exchange protocol contribute to the establishment of secure communication channels in modern cryptographic systems, and what are the underlying mathematical principles that ensure its effectiveness?	6	4	4	2,3,6,8	

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Department of Computer Science and Engineering

MID SEMESTER EXAMINATION-I
Academic Year 2023-24 Semester-II

Date: 24/02/2024
Time: 1 hr

Class: TY CSE
Course: CNS
Max Marks: 15

Q. 1	Solve/Answer the following questions	Marks	CO	BL	PO	PI
	Questions					
a)	What is the difference between symmetric and asymmetric cryptography?	2	1	2	1,2,8	
b)	Using Euclidean Algorithm, GCD (21,45) =?	2	2	2	2,3,6,8	
Q. 2	Solve/Answer Any one					
a)	Compare Active Attacks and Passive Attacks	5	1	3	1,2,8	
b)	With a neat diagram explain the Model for network security	5	1	3	1,2,8	
Q. 3	Solve/Answer Any one					
a)	You are working as a cryptographer for a confidential communication system. Your team has decided to use a polyalphabetic cipher to encrypt sensitive messages. You are given the following text to encrypt using a Vigenère cipher. Provide the step-by-step process and show the resulting encrypted text. Plain text: "ATTACKCANCELLED" Key: "MYKEY"	6	2	3	2,3,6,8	
b)	As a communications specialist working for a covert intelligence agency, you need to encrypt a sensitive message before sending it over an insecure channel. You decide to use a row transposition cipher for added security. The original message you need to encrypt is: MEET ME AT STATION. use the key "2 1 4 3." Provide the step-by-step process and show the resulting encrypted text.	6	2	3	2,3,6,8	

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END SEMESTER EXAMINATION - June/July-2023-24

B.TECH VI Semester (CSE)

Course Code: CSE351

Duration: 2 Hours

Course Name: Cryptography and Network Security

Max. Marks: 50

Date: 01 June 2024

Important Instructions:

1. All questions are compulsory.
2. Assume suitable data wherever necessary and clearly state it.
3. Figures to right indicate full marks.

01 JUN 2024

Question 1: Answer any 5 questions.

	Marks CO BL		
a. Distinguish between stream cipher and block cipher.	2	3	2
b. What is man in the middle attack?	2	1	2
c. Encrypt the following message using rail fence cipher. Message : "This is era of artificial intelligence"	2	2	3
d. Describe the asymmetric key cryptography? Which keys are used by the sender in asymmetric cryptography for (a) Encryption (b) Signature	2	2	3
e. Compare active attacks and passive attacks	2	1	2
f. Describe Euclidean algorithm, calculate the GCD(1970,1066)	2	1	3
g. In the context of digital certificate, What is the role of CA ?	2	6	2
h. What significant properties are required for a hash function to be useful for message authentication?	2	5	2

Question 2: Answer any 2 questions.

	Marks CO BL		
a. Provide an example illustrating the transformation of plain text into cipher text using a transposition encryption algorithm	4	2	2
b. Analyze the key differences between a chosen-ciphertext attack and a chosen-plaintext attack, and discuss their implications on cryptographic security.	4	2	3
c. Compare and contrast substitution and transposition techniques in cryptography, analyzing their respective strengths and weaknesses. Provide examples to illustrate the application of both substitution and transposition techniques in cryptographic systems.	4	2	2

Question 3: Answer any 2 questions.

	Marks CO BL		
a. Compare and contrast at least three different modes of operation in symmetric key cryptography, highlighting their respective strengths and weaknesses	4	3	2
b. Provide a detailed explanation of each transformation function used in AES, including SubBytes, ShiftRows, MixColumns, and AddRoundKey	4	3	2
c. Discuss the strengths and weaknesses of Blowfish, as well as its suitability for various cryptographic applications	4	3	2

Question 4: Answer any 2 questions.

	Marks CO BL		
a. Explain the primary function of a digital certificate in online communications and analyze its impact on ensuring secure interactions. Provide examples to illustrate your explanation.	4	6	3
b. Explain how a digital signature ensures the integrity of a message and analyze its role in maintaining data security. Provide an example to support your explanation	4	6	3
c. Identify and explain common threats in network security. Analyze the potential impacts of these threats on an organization's data and operations.	4	6	3

Question 5: Answer any 1 questions.

	Marks CO BL		
a. In an RSA system, the public key of a given user is $(7, n = 137)$. What is the private key of this user?	8	4	3
b. Analyze the potential vulnerabilities in the Diffie-Hellman key exchange protocol. Describe how these vulnerabilities can be exploited by an attacker and propose possible mitigations for the vulnerability.	8	4	4

Question 6: Answer any 1 questions.

Marks CO BL

- a. Discuss the security requirements necessary for generating message digests using the SHA-512 cryptographic hash function. Analyze the implications of these requirements on ensuring the integrity data. 8 5 3
- b. Examine the essential requirements for ensuring the security of Message Authentication Codes (MACs) in cryptographic systems. Analyze potential cryptanalysis techniques that could compromise the security of MACs. 8 5 3

—End of question paper—

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Department of Computer Science & Engg.

MID SEMESTER EXAMINATION-I
Academic Year 2023-24 Semester-II

Class: TY (CSE)
Course: Machine Learning
Max Marks: 15

Date:
Time: 1 hr

Q. 1	Solve/Answer the following questions																																																																	
	Questions	Marks	CO	BL	PO	PI																																																												
a)	What are the Machine Learning life cycle stages?	2	1	1	1																																																													
b)	Explain Bayesian Theorem.	2	2	2	1																																																													
Q. 2	Solve/Answer Any one																																																																	
a)	Explain with diagram underfitting and overfitting of a model.	5	1	2	1																																																													
b)	What is the difference between supervised and unsupervised learning? Give example of each.	5	1	2	1																																																													
Q. 3	Solve/Answer Any one																																																																	
a)	Of three persons the chances that a politician, a businessman or an academician would be appointed the Vice Chancellor (VC) of a university are 0.5, 0.3, 0.2 respectively. Probabilities that research will be promoted by these persons if appointed are 0.3, 0.7 and 0.8 respectively. Find the probability that if research is promoted, (i) VC is an academician (ii) VC is a businessman.	6	2	5	4																																																													
b)	Consider the following employee dataset. The data have been generalized. For example, "31 ... 35" for age represents the age range of 31 to 35. Classify the following sample using Bayesian Classification: X = (Department = "systems," Age = "26--30," and Salary = "46--50K") <table><tr><th>Sr. No</th><th>Department</th><th>Age</th><th>Salary</th><th>Class: Status</th></tr><tr><td>1</td><td>Sales</td><td>31--35</td><td>46K--50K</td><td>Senior</td></tr><tr><td>2</td><td>Sales</td><td>26--30</td><td>26K--30K</td><td>Junior</td></tr><tr><td>3</td><td>Sales</td><td>31--35</td><td>31K--35K</td><td>Junior</td></tr><tr><td>4</td><td>Systems</td><td>21--25</td><td>46K--50K</td><td>Junior</td></tr><tr><td>5</td><td>Systems</td><td>31--35</td><td>66K--70K</td><td>Senior</td></tr><tr><td>6</td><td>Systems</td><td>26--30</td><td>46K--50K</td><td>Junior</td></tr><tr><td>7</td><td>Systems</td><td>41--45</td><td>66K--70K</td><td>Senior</td></tr><tr><td>8</td><td>Marketing</td><td>36--40</td><td>46K--50K</td><td>Senior</td></tr><tr><td>9</td><td>Marketing</td><td>31--35</td><td>41K--45K</td><td>Junior</td></tr><tr><td>10</td><td>Secretary</td><td>46--50</td><td>36K--40K</td><td>Senior</td></tr><tr><td>11</td><td>Secretary</td><td>26--30</td><td>26K--30K</td><td>Junior</td></tr></table>	Sr. No	Department	Age	Salary	Class: Status	1	Sales	31--35	46K--50K	Senior	2	Sales	26--30	26K--30K	Junior	3	Sales	31--35	31K--35K	Junior	4	Systems	21--25	46K--50K	Junior	5	Systems	31--35	66K--70K	Senior	6	Systems	26--30	46K--50K	Junior	7	Systems	41--45	66K--70K	Senior	8	Marketing	36--40	46K--50K	Senior	9	Marketing	31--35	41K--45K	Junior	10	Secretary	46--50	36K--40K	Senior	11	Secretary	26--30	26K--30K	Junior	6	2	5	4	
Sr. No	Department	Age	Salary	Class: Status																																																														
1	Sales	31--35	46K--50K	Senior																																																														
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6	Systems	26--30	46K--50K	Junior																																																														
7	Systems	41--45	66K--70K	Senior																																																														
8	Marketing	36--40	46K--50K	Senior																																																														
9	Marketing	31--35	41K--45K	Junior																																																														
10	Secretary	46--50	36K--40K	Senior																																																														
11	Secretary	26--30	26K--30K	Junior																																																														

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Department of Computer Science & Engineering

MID SEMESTER EXAMINATION-II
 Academic Year 2023-24 Semester-II

Class: TY(CSE)

Course: Machine Learning

Max Marks: 15

Date:

Time: 1 hr

Q. 1 Solve/Answer the following questions																																																							
	Questions	Marks	CO	BL	PO	PI																																																	
a)	Explain in short SVM.	2	3	2	1																																																		
b)	Write the steps for clustering using K-Means clustering.	2	4	2	1																																																		
Q. 2 Solve/Answer Any one																																																							
a)	What is K-NN approach for problem solving? Apply K-NN with K = 3 for classification of test sample: (Height, weight) = (170, 57) as belonging to under weight or normal weight category. <table><tr><th>Height (CM)</th><th>Weight (KG)</th><th>Classification</th></tr><tr><td>167</td><td>51</td><td>Under weight</td></tr><tr><td>182</td><td>62</td><td>Normal</td></tr><tr><td>176</td><td>69</td><td>Normal</td></tr><tr><td>173</td><td>64</td><td>Normal</td></tr><tr><td>172</td><td>65</td><td>Normal</td></tr><tr><td>174</td><td>56</td><td>Under weight</td></tr><tr><td>169</td><td>58</td><td>Normal</td></tr><tr><td>173</td><td>57</td><td>Normal</td></tr><tr><td>170</td><td>55</td><td>Normal</td></tr></table>	Height (CM)	Weight (KG)	Classification	167	51	Under weight	182	62	Normal	176	69	Normal	173	64	Normal	172	65	Normal	174	56	Under weight	169	58	Normal	173	57	Normal	170	55	Normal	5	3	3	1																				
Height (CM)	Weight (KG)	Classification																																																					
167	51	Under weight																																																					
182	62	Normal																																																					
176	69	Normal																																																					
173	64	Normal																																																					
172	65	Normal																																																					
174	56	Under weight																																																					
169	58	Normal																																																					
173	57	Normal																																																					
170	55	Normal																																																					
b)	Explain ID3 decision tree algorithm with example.	5	3	3	1																																																		
Q. 3 Solve/Answer Any one																																																							
a)	Consider the following dataset of 6 samples and apply K-Means clustering to create 2 clusters (K=2) from this dataset with Euclidean distance. Consider initial centroids c1 (1,4) and c2(5,1). Sample points (A1 to A6): (1, 4), (1, 3), (0, 4), (5, 1), (6, 2) and (4, 0)	6	4	3	4																																																		
b)	Consider the following distance matrix and solve using agglomerative hierarchical clustering - Complete Linkage and draw a dendrogram: <table><tr><th></th><th>P1</th><th>P2</th><th>P3</th><th>P4</th><th>P5</th><th>P6</th></tr><tr><td>P1</td><td>0</td><td>0.71</td><td>5.66</td><td>3.6</td><td>4.24</td><td>3.20</td></tr><tr><td>P2</td><td>0.71</td><td>0</td><td>4.95</td><td>2.92</td><td>3.53</td><td>2.5</td></tr><tr><td>P3</td><td>5.66</td><td>4.95</td><td>0</td><td>2.24</td><td>1.41</td><td>2.5</td></tr><tr><td>P4</td><td>3.6</td><td>2.92</td><td>2.24</td><td>0</td><td>1.0</td><td>0.5</td></tr><tr><td>P5</td><td>4.24</td><td>3.53</td><td>1.41</td><td>1.0</td><td>0</td><td>1.12</td></tr><tr><td>P6</td><td>3.20</td><td>2.5</td><td>2.5</td><td>0.5</td><td>1.12</td><td>0</td></tr></table>		P1	P2	P3	P4	P5	P6	P1	0	0.71	5.66	3.6	4.24	3.20	P2	0.71	0	4.95	2.92	3.53	2.5	P3	5.66	4.95	0	2.24	1.41	2.5	P4	3.6	2.92	2.24	0	1.0	0.5	P5	4.24	3.53	1.41	1.0	0	1.12	P6	3.20	2.5	2.5	0.5	1.12	0	6	4	3	4	
	P1	P2	P3	P4	P5	P6																																																	
P1	0	0.71	5.66	3.6	4.24	3.20																																																	
P2	0.71	0	4.95	2.92	3.53	2.5																																																	
P3	5.66	4.95	0	2.24	1.41	2.5																																																	
P4	3.6	2.92	2.24	0	1.0	0.5																																																	
P5	4.24	3.53	1.41	1.0	0	1.12																																																	
P6	3.20	2.5	2.5	0.5	1.12	0																																																	

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END SEMESTER EXAMINATION - June/July-2023-24

B.TECH VI Semester (CSE)

Course Code: CSE353

Duration: 2 Hours

Max. Marks: 50

Course Name: Machine Learning

Date: 06 June 2024

Important Instructions:

1. All questions are compulsory.
2. Assume suitable data wherever necessary and clearly state it.
3. Figures to right indicate full marks.

Question 1: Answer any 5 questions.

- a. What are the life cycle stages of Machine Learning.
- b. Explain the concept of Hypothesis.
- c. What is K-Fold Cross Validation?
- d. Differentiate between Linear and Logistic Regression.
- e. Write the steps of KNN based algorithm.
- f. Enlist any four linkages used in Hierarchical clustering.
- g. Explain the basic architecture of a neuron in short.
- h. Enlist any four techniques based on Ensemble Learning

Marks CO BL

2	1	1
2	1	2
2	1	2
2	2	2
2	3	2
2	4	1
2	6	2
2	5	1

Question 2: Answer any 2 questions.

- a. Write short note on SVM and SVM Kernel functions.
- b. Apply K-NN with $K = 3$ for classification of test sample: $(X_1, X_2) = (3, 7)$ as Good or Bad. Consider the following dataset and use Euclidean distance formula:

Marks CO BL

4	3	2
4	3	3

Sample. No.	X1	X2	Classification
1	7	7	Bad
2	7	4	Bad
3	3	4	Good
4	1	4	Good

- c. Consider the following dataset of House Price Index (HPI). Age and Loan are independent variable and HPI is the value to be predicted.

4 3 3

Sample No.	Age	Loan	House Price Index
1	25	40000	135
2	35	60000	256
3	45	80000	231
4	20	20000	267
5	35	120000	139

Use KNN and consider $K = 3$ to predict HPI for a person with Age = 40 and Loan = 75000

Question 3: Answer any 2 questions.

- a. Explain ensemble technique with Averaging.
- b. Write short note on ensemble learning through Boosting.
- c. Write short note on ensemble learning through Stacking.

Marks CO BL

4	5	2
4	5	2
4	5	2

Question 4: Answer any 2 questions.

- a. Explain any two activation function used in neural networks.

Marks CO BL

4	6	2
---	---	---

- b. Write short note on Autoencoders.
c. Explain backpropagation in brief.

4 6 2
4 6 2

Question 5: Answer any 1 questions.

Marks CO BL

- a. What is Bayesian Classification? Solve the following problem using Bayesian Theorem.
In a bolt factory, machines A, B and C manufacture 25%, 35% and 40% of the total output. Out of the total manufacturing, 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the products and found to be defective. Find the probability that it is manufactured from (i) Machine A (ii) Machine B and (iii) Machine C
- b. Classify the following sample using Bayes classification as Accident = Yes or No:
X = (Weather = "Rain", Road = "Good", Traffic = "High", Engine Problem = "No")
Consider following dataset.

8 2 5

8 2 5

Sample No.	Weather	Road	Traffic	Engine Problem	Class: Accident
1	Rain	Bad	High	No	Yes
2	Snow	Average	Normal	Yes	Yes
3	Clear	Bad	Light	No	No
4	Clear	Good	Light	Yes	Yes
5	Snow	Good	Normal	No	No
6	Rain	Average	Light	No	No
7	Rain	Good	Normal	No	No
8	Snow	Bad	High	No	Yes
9	Clear	Good	High	Yes	No
10	Clear	Bad	High	Yes	Yes

Question 6: Answer any 1 questions.

Marks CO BL

- a. Consider the following dataset and compute 2 clusters (K = 2) with initial centroids c1 (1.0, 1.0) and c2 (5.0, 7.0) using K-Means clustering. Calculate up to 2 iterations.

8 4 3

Sample No	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

- b. Consider the following distance matrix and solve using agglomerative hierarchical clustering – complete Linkage and draw the dendrogram:

8 4 3

	P1	P2	P3	P4	P5
P1	0.0	0.10	0.41	0.55	0.35
P2	0.10	0.0	0.64	0.47	0.98
P3	0.41	0.64	0.0	0.44	0.85
P4	0.55	0.47	0.44	0.0	0.76
P5	0.35	0.98	0.85	0.76	0.0

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Department of Emerging Science And Technology						

MID SEMESTER EXAMINATION-II
Academic Year 2023-24 Semester-I

Class: TY

Cours: Digital Marketing (OE-1) AIDS/Agri/ECE/EEE/MECH/PPE Date: 10/04/2024

Max Marks: 15

Time: 1 hr

Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	What do you mean by smo	2	3	1	1	1.3.1
b)	Explain with example ' need of Facebook profile creation for post and promotion '.	2	3	3	1	1.3.1
Q. 2	Solve/Answer Any one					
a)	How to analyzed digital marketing analytics	5	3	3	1	1.3.1
b)	Write in short about Google packages for advertising	5	4	2	2	2.1.3
Q. 3	Solve/Answer Any one					
a)	How to generate a report using Google AdWords or AdSense	6	4	2	2	2.1.3
b)	What do you mean by mapping ads	6	4	3	2	2.1.3

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Department of _____

MID SEMESTER EXAMINATION-I
Academic Year 2023-24 Semester-II

Date: _____

Time: 1 hr

Class: TYCSE

Course: OE-III DM

Max Marks: 15

Solve/Answer the following questions		Marks	CO	BL	PO	PI
Q. 1	Questions					
a)	Define digital marketing	2	1	2	5	2.1.1
b)	What are the types of digital marketing	2	1	2	5	2.1.2
Q. 2	Solve/Answer Any one					
a)	Apply digital marketing strategies for advertising of product.	5	1	3	6	3.1.1
b)	Identify the difference between traditional and digital marketing.	5	1	3	6	3.1.2
Q. 3	Solve/Answer Any one					
a)	Apply competition analysis on product data	6	2	3	6	3.1.1
b)	What is SEO? Describe Search engine optimization working in detail.	6	2	2	6	2.1.2

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END SEMESTER EXAMINATION - June/July-2023-24

B.TECH VI Semester (CSE)

Course Code: CSE392

Course Name: Open EL-III Digital Marketing

Duration: 2 Hours

Max. Marks: 50

Date: 10 June 2024

Important Instructions:

1. All questions are compulsory.
2. Assume suitable data wherever necessary and clearly state it.
3. Figures to right indicate full marks.

Question 1: Answer any 5 questions.

- a. Define Digital Marketing strategy.
- b. What are the types of Digital Marketing.
- c. What is ROI between Digital and Traditional Marketing.
- d. How to use Google Ad words Express.
- e. How to create Facebook Profile.
- f. What is Search engine optimization
- g. What is Email Marketing.
- h. Enlist the types of Advertisement.

Marks CO BL

2	1	2
2	1	2
2	1	1
2	4	2
2	3	2
2	2	1
2	6	2
2	5	2

Question 2: Answer any 2 questions.

- a. Discuss Google Algorithm and its updates in detail.
- b. Differentiate Between Web position Analysis and Competition Analysis.
- c. Discuss the importance of user Experience (UX) factors in search engine ranking.

Marks CO BL

4	2	3
4	2	3
4	2	3

Question 3: Answer any 2 questions.

- a. Apply promotional technique to do LinkedIn post and promotions using digital marketing.
- b. Discuss Integration technique of Instagram in detail.
- c. Discuss the key differences between social media optimization and search engine optimization

Marks CO BL

4	3	3
4	3	3
4	3	3

Question 4: Answer any 2 questions.

- a. Illustrate the concept of mobile Advertising in detail.
- b. Describe in detail Google mapping ads.
- c. Discuss the different types of video ads formats available for online advertising

Marks CO BL

4	4	3
4	4	3
4	4	3

Question 5: Answer any 1 questions.

- a. Write a short note on E-Commerce Management.
- b. Describe in detail the concept of customer retention.

Marks CO BL

8	5	2
8	5	2

Question 6: Answer any 1 questions.

- a. Explain in detail conversion rate of emails.
- b. Discuss in detail what is product based email with example.

Marks CO BL

8	6	3
8	6	3

—End of question paper—

G. S. Mandal's
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Department of Computer Science and Engineering

MID SEMESTER EXAMINATION-II
Academic Year 2023-24 Semester-II

Class: TYCSE

Date: 06/04/2024

Course: Principles of Compiler Design

Time: 1 hr

Max Marks: 15

Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	Differentiate CLR and SLR Parser	2	2	2	1	1.6.1
b)	Explain types of Attributes in SDT with suitable example	2	3	2	1	1.6.1
Q. 2	Solve/Answer Any one					
a)	Consider the following grammar and their syntax Directed Translation (SDT) rules $S \rightarrow S * A \quad \{S.val = S.val \times A.val\}$ $S \rightarrow A \quad \{S.val = A.val\}$ $A \rightarrow A + B \quad \{A.val = A.val + B.val\}$ $A \rightarrow B \quad \{A.val = B.val\}$ $B \rightarrow num \quad \{B.val = num.val\}$ Derive the tree and solve the expression $7+9*6+10$	5	3	3	1,2	2.5.3
b)	Illustrate SDT for conversion of infix to postfix expression with suitable Example	5	3	3	1,2	2.5.3
Q. 3	Solve/Answer Any one					
a)	Consider the grammar $S \rightarrow AA$ $A \rightarrow 0A \mid 1$ Construct LALR Parsing table and show the steps for acceptance of the string 011\$	6	2	3	1,2	2.5.3
b)	Consider the grammar and apply SLR Parser $S \rightarrow L = R \mid R$ $L \rightarrow * R \mid id$ $R \rightarrow L$	6	2	3	1,2	2.5.3

Student Copy

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Department of Computer Science and Engineering

MID SEMESTER EXAMINATION-I

Academic Year 2023-24 Semester-II

Class: TYCSE

Date: 26/02/2024

Course: Principles of compiler design

Time: 1 hr

Max Marks: 15

Q. 1	Solve/Answer the following questions					
	Questions	Marks	CO	BL	PO	PI
a)	Enlist compiler writing tools , Write structure of Lex Program	2	1	1	1	1.4.3
b)	Find FIRST and FOLLOW set of $S \rightarrow AaAb / BbBa$ $A \rightarrow \epsilon$ $B \rightarrow \epsilon$	2	2	2	1	1.4.3
Q. 2	Solve/Answer Any one					
a)	Design and implement lexical analyzer using regular expression and transition diagram of token identifier.	5	1	3	1	1.4.3
b)	Convert Regular Expression $a^*.b^*.c^*$ to equivalent FA	5	1	3	1	1.4.3
Q. 3	Solve/Answer Any one					
a)	Explain operating precedence parser for the grammar $E \rightarrow E+T/T$ $T \rightarrow T*F/F$ $F \rightarrow id$ Process the string $id*id+id\$$	6	2	3	1	1.4.3
b)	Apply Recursive Descent Parser algorithm on given grammar $E \rightarrow TE'$ $E' \rightarrow +TE' e$ $T \rightarrow FT'$ $T' \rightarrow *FT' e$ $F \rightarrow (E) id$	6	2	3	1	1.4.3